QU4TTRO

BY VRCVS

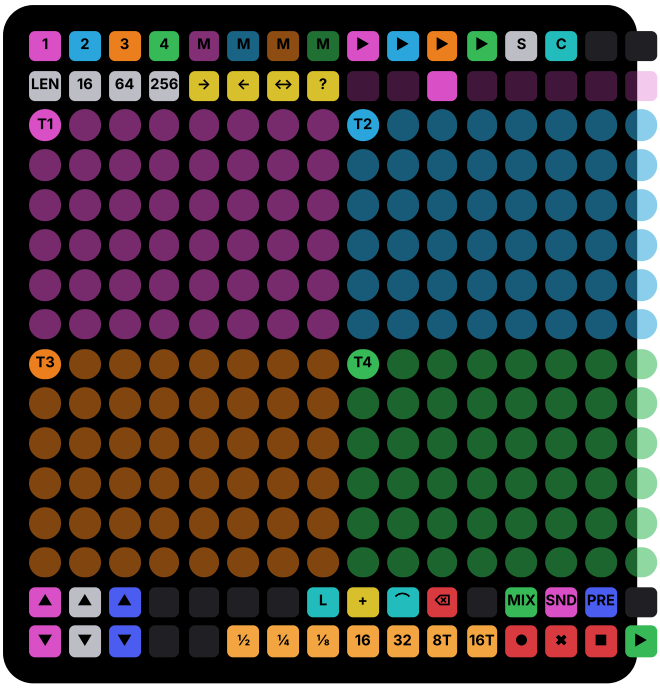
Custom panel for Plinky 12 · Chords faceplate

QU4TTRO turns Plinky 12 into four melodic instruments side by side, each with its own sound and its own loop. It plays like plonky: every column is a string, and the column to the right starts a fifth higher. Loops can be recorded live or step by step.

CREDITS

QU4TTRO is a port for Plinky 12, freely inspired by the norns script **plonky** by **Zack Scholl** (schollz / infinitedigits), github.com/schollz/plonky, MIT licence. plonky was itself inspired by mmalex's original Plinky. Plinky 12 is by mmalex and Making Sound Machines. QU4TTRO panel by VRCVS (Filippo Rieder), written with help from Claude.

1 · Panel map



The panel has 16×16 pads: 2 rows of rectangles at the top, 12 rows of round pads in the middle (the 4 tracks), and triangles and rectangles with the transport at the bottom. Every control except the transport acts on the selected track (rows 1–2 and the row above the bottom one). Octave, key and scale have one colour per value, which stays lit so you can read it at a glance. They flash white for a moment on every change. Brightness: **very bright** = active or selected, **lit** = available, **dim** = off or not selected.

1 2 3 4	Row 1: select track T1–T4
M	Row 1: mute tracks T1–T4
▶ ▶ ▶ ▶	Row 1: play/stop for each track (blinks in time while running)
S	Row 1: start mode: white = from step 1, yellow = resume, cyan = from the CUE step
C	Row 1: CUE: hold it and tap a pad of the track to choose the start step
LEN	Row 2: hold it and tap a pad of the track to set the length (1–48 steps)
16 64 256	Row 2: quick lengths in steps (16 = 1 bar at 1/16, 256 = 16 bars)
→ ← ↔ ?	Row 2: direction: forward, reverse, ping-pong, random
▮	Row 2, right: loop position, or the step you are writing
▲▼ (1st)	Track octave: purple −3, red −2, orange −1, white 0, cyan +1, green +2, yellow +3
▲▼ (2nd)	Key: C red, D orange, E yellow, F green, G cyan, A blue, B purple; sharps take the colour of the note below, dimmer
▲▼ (3rd)	Scale: minor green, dorian cyan, phrygian purple, minor pentatonic teal, major blue, lydian yellow, hirajoshi pink, chromatic white
L	Mono legato on/off for the selected track
+	Step rec: +STEP. With notes held it stretches them by 1 step; with nothing held it adds 1 silent step
⁀	Step rec: next note is entered legato
⌫	Step rec: back (delete the last step)
½ … 16T	Bottom row: grid of the selected track: 1/2, 1/4, 1/8, 1/16, 1/32, 1/8 triplet, 1/16 triplet
MIX	Mixer: volume, pan, delay and reverb for the 4 tracks
SND	Sound of the selected track
PRE	Presets: sound picker (instant preview); ● = alternative SD browser
●	Rec. Playing, empty track: arms, the loop starts on the first note; tap again = close the loop; on a loop = overdub. Stopped: step entry
✖	Held while playing: eraser. Held 1 s while stopped: clear the track
■ ▶	Stop · Play

The four tracks

T1 top left · **T2** top right · **T3** bottom left (one octave down: the bass) · **T4** bottom right. Each sector is 8×6: every **column is a string**, and the note rises along the scale as you go up; the column to the right starts **a fifth higher**. Brighter pads are root notes; fully lit pads are sounding. **Pressure** = dynamics, even while you hold the note. **Sliding** your finger from pad to pad without lifting moves to the next note without retriggering (glide, if the preset has GLIDE). Plinky plays up to 8 notes at once in total.



One sector in D minor: numbers are scale degrees (1 = root).

2 · Give each track a sound

1. Tap the 1st rectangle at the top to select T1.
2. Press **PRE** (row above the bottom one, above ✖): the Plinky **sound picker** opens, the same as on the factory panels, moved onto the round pads. Banks at the top, categories below: tap a bank, then a preset to **hear it straight away**.
3. Press ▶ to load it onto the selected track and go back to playing (big multisample banks can take a while to load). ✖ cancels and goes back to playing.
4. On row 14 the first 4 pads select the track, so you can load all four without leaving. ● opens the alternative SD browser (see section 9); there, ● takes you back to the picker.
5. **SND** (above ■) edits the sound of the selected track. **MIX** (above ●) opens the mixer for the 4 tracks (see section 7). In both, the bottom-right pad takes you back to playing.

3 · Recording live

1. Select a track and choose its grid in the bottom row (1/2 … 1/32, triplets). Notes are quantised to that grid.
2. Press ▶ play.
3. Tap ●: it blinks, the track is armed. Wait as long as you like: the loop starts with the first note you play.
4. Tap ● again to close the loop, like a looper: the length runs from the first note to the step where you press it (trailing rests included), and the loop restarts from step 1 right after, in time.
5. A loop can last up to 256 steps: 16 bars at 1/16, 32 at 1/8, 64 at 1/4. If you reach the limit without closing it, the loop wraps to step 1 and carries on overdubbing.
6. Tap ● again to overdub: add notes or polyphonic voices inside the loop, which keeps its length. Tap ● to stop: the loop carries on as normal.
7. Every track has its own loop with its own length: record them one after another while the others play.
8. To fix mistakes: while playing, hold ✖ to erase as the loop passes. To start a track over: while stopped, hold ✖ for 1 second.

4 · Step recording (plonky style)

With the transport stopped (■), tap ●. Every note you press and release writes one step and moves to the next. The loop lasts up to the last step written.

A note	Tap and release a pad.
A chord	Press several pads together, then release them.
A silent step	With no notes held, press **+** (+STEP).
Make a note or chord last longer	Keep it held and press **+**: each press stretches it by 1 step. Then release.
Connect two notes	⁀ before the next note. Best heard with mono legato (L) on and some GLIDE.
Fix the last step	⌫ back
Start over	While stopped, hold ✖ for 1 second.

The step you are writing shows on the right of row 2: the lit pad is the step within a group of 8, and the colour switches between red and white for each group. Each step lasts one grid unit of the track (with 1/8, every step is an eighth note).

Example



Hold the chord and press + twice (a 3-step chord), release, press + with nothing held (rest), then note, ⁀ and note. Press ■, turn ● off, then ▶ to listen.

5 · Each track's loop

- **Different lengths** per track, 1 to 256 steps (16 bars at 1/16, 64 at 1/4), create polyrhythms: with T1 = 16, T2 = 12, T3 = 7 and T4 = 5 the loops only line up again after many passes.
- **Grid** (bottom row): 1/2, 1/4, 1/8, 1/16, 1/32 and triplets, per track. It sets how long a step lasts and where notes are quantised. Different grids = more polyrhythms (e.g. T1 at 1/16, T2 in triplets, T3 at 1/4).
- **Direction**: forward, reverse, ping-pong or random, per track.
- **Mute**: silences the loop; you can still play the track live.
- **Mono legato (L)**: one note at a time, moving to the next without retriggering. With high GLIDE notes slide.
- **BPM**: left side buttons.

6 · Starting tracks independently

- ▶ and ■ at the bottom are the global transport and keep time for everything. ▶ starts only the tracks that are switched on.
- While playing, switch each track on or off with its play/stop (row 1, rectangles 9–12). It comes in on time, on the next 16th.
- Where it starts from depends on its **start mode** (rectangle 13): from step 1, from where it stopped, or from the CUE step.
- To choose the CUE step, hold rectangle 14 and tap a pad of the track: the pad lights cyan and the mode switches to CUE automatically.
- Selector pads (LEN and CUE) count left to right, row by row from the top: the top-left pad is step 1.
- A track that is switched off doesn't play its loop, but you can still play it live.

7 · Mixer, effects and sound

- **MIX · Mixer.** 4 columns per track, T1–T4 from the left: **volume · pan/stereo · delay send · reverb send**. Under each group: 2 pads to select the track and 2 to mute it.
- **SND · Sound** of the selected track. Bottom block (16 sliders): octave, pitch, glide, sample start, length, timestretch, chorus, sub-oscillator, wavefold, high-pass, low-pass, resonance, attack, decay, sustain, release.
- Top block: the track's volume, stereo, delay send, reverb send and toad factor, then the **global effects**: audio input (dry, delay, reverb), delay time and feedback, reverb shimmer and feedback, bass/mid/treble EQ, output level. Delay, reverb and EQ are shared by all tracks: each track decides how much it sends.
- Below the sliders are the sound-mode buttons (simple, tune, chop, loop, sync, low-pass gate, fixed pitch) and the XY pad.
- **Pressure as modulation:** in SND you can assign the **P** (pressure) mod source to a parameter, e.g. the filter. It works live and on recorded notes, since QU4TTRO records pressure step by step.

8 · Saving and loading

- **Song** (right side button ▼). Saves or loads a song: every track's loop with notes, ties and pressure, plus lengths, grids, directions, octaves, mutes, play/stop, start modes and CUE, scale and key, the 4 sounds and the mix with its effects.
- Pick a slot on the right (folders on the left) and confirm with the bottom-right pad. The pad next to it cancels.
- The last song you loaded or saved is reloaded automatically at power-on.
- **PRE** loads the sound of the selected track from the SD browser.
- **Starting QU4TTRO:** the IDE can keep up to 5 panels open in separate tabs, and Flash installs **all of them together**, next to the factory panels. At power-on, hold the bottom-row pad of the panel you want (the diagram under the IDE emulator shows which) and choose QU4TTRO. To go back to the factory Chords, power on holding the 2nd pad from the left on the bottom row (1st = Blocks, 3rd = Toadstep).

9 · Alternative SD browser

- In PRE, ● opens an alternative browser that reads the SD folders directly: row 3 = categories, left (8×8) the banks, right (8×8) the presets in the file grid. Tapping a preset loads it for audition (slower than the picker), ▶ puts it on the track instantly, ✖ cancels.
- QU4TTRO reads the folders inside **PLINKY/BANKS** by itself: each folder is a **category**, each sub-folder a Plinky **bank** (for example one made with the Bank Builder).
- Names are free (e.g. VRCVS_INSTR/DUDUK, MY_PIANOS/SOFT_PIANO), up to **16 characters**. They appear in alphabetical order.
- Limits: **16 categories** in total (including the 7 factory ones, so 9 of yours), **64 banks** per category, **64 presets** per bank (8×8 grid, files 00.json to 77.json).
- The list refreshes every time you open PRE.
- A folder with a name that is too long is skipped. Presets are XY.json files: X = row, Y = column, digits 0 to 7 (after 07 comes 10).
- If a Plinky cannot list folders, PRE falls back to a fixed layout: PLINKY/BANKS/USER1 and USER2, with banks named BANK1 … BANK8.

10 · Large samples and memory

- To play a sample, the Plinky copies it from the SD into its internal memory, which is limited. QU4TTRO keeps up to **4 sounds in memory at once** (one per track), plus the one being auditioned.
- Large multisample banks (e.g. an 8 MB tape.bin) fill most of that memory on their own: loading with ▶ can take **over a minute**, and a second large bank on another track may **fail to load** (the load stops and is cancelled).
- Factory sounds are small and load instantly; you can mix them freely with yours.
- To use several of your own sounds together, make lighter banks: roughly **1–2 MB** per bank. In the Bank Builder: fewer sampled notes (the Plinky pitches the ones in between), shorter samples with a loop for the sustain, mono and a lower sample rate if available.
- The preview in the picker is always instant: the wait only applies to loading onto the track.

Navigation: MIX, SND and PRE open mixer, sound and presets; the bottom-right pad goes back to playing. The right side button ▼ opens song saving, ▲ goes back. If one of your banks won't load on a second track, see section 10.